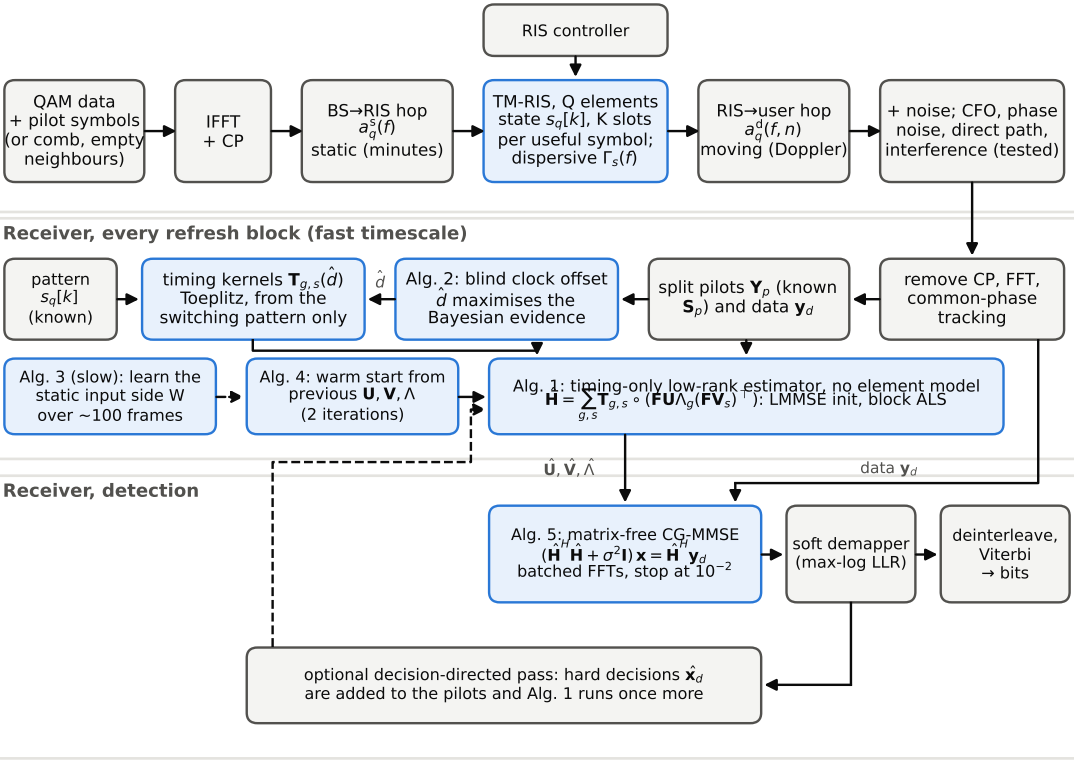
Transmitter and channel
RIS controller
QAM data + pilot symbols (or comb, empty neighbours)
IFFT + CP
BS→RIS hop $a_q^s(f)$ static (minutes)
TM-RIS, Q elements state $s_q[k]$, K slots per useful symbol; dispersive $\Gamma_s(f)$
RIS→user hop $a_q^d(f,n)$ moving (Doppler)
+ noise; CFO, phase noise, direct path, interference (tested)
Receiver, every refresh block (fast timescale)
pattern $s_q[k]$ (known)
timing kernels $\mathbf{T}_{g,s}(\hat{d})$ Toeplitz, from the switching pattern only
$\hat{d}$
Alg. 2: blind clock offset $\hat{d}$ maximises the Bayesian evidence
split pilots $\mathbf{Y}_p$ (known $\mathbf{S}_p$) and data $\mathbf{y}_d$
remove CP, FFT, common-phase tracking
Alg. 3 (slow): learn the static input side W over ~100 frames
Alg. 4: warm start from previous $\mathbf{U}, \mathbf{V}, \Lambda$ (2 iterations)
Alg. 1: timing-only low-rank estimator, no element model $\mathbf{H} = \sum_{g,s} \mathbf{T}_{g,s} \circ (\mathbf{F}\mathbf{U}\Lambda_g(\mathbf{F}\mathbf{V}_s)^\top)$: LMMSE init, block ALS
$\hat{\mathbf{U}}, \hat{\mathbf{V}}, \hat{\Lambda}$
data $\mathbf{y}_d$
Receiver, detection
Alg. 5: matrix-free CG-MMSE $(\hat{\mathbf{H}}^H\hat{\mathbf{H}} + \sigma^2\mathbf{I})\mathbf{x} = \hat{\mathbf{H}}^H\mathbf{y}_d$ batched FFTs, stop at 10^{-2}
soft demapper (max-log LLR)
deinterleave, Viterbi → bits
optional decision-directed pass: hard decisions $\hat{\mathbf{x}}_d$ are added to the pilots and Alg. 1 runs once more